

CHIFENG, China

WMO: 542180

Lat: 42.3080N

Lon: 118.8325E

Elev: 2152

StdP: 13.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.0	(c) -1.6	(d) -23.4	(e) 1.6	(f) 4.1	(g) -19.9	(h) 2.0	(i) 5.8	(j) 20.5	(k) 19.2	(l) 18.2	(m) 19.7	(n) 2.8	(o) 220	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.3	(c) 91.6	(d) 67.2	(e) 88.6	(f) 66.6	(g) 86.0	(h) 65.6	(i) 73.2	(j) 84.4	(k) 71.5	(l) 81.6	(m) 70.0	(n) 79.8	(o) 7.0	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	119.3	79.0	68.3	112.7	77.4	66.6	106.2	75.9	38.5	84.2	36.9	81.6	35.5	79.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.8	(b) 16.2	(c) 13.9	(e) -9.0	(f) 96.9	(g) 4.1	(h) 3.3	(i) -11.9	(j) 99.2	(k) -14.3	(l) 101.1	(m) -16.6	(n) 102.9	(o) -19.6	(p) 105.3		
(a) 18.8	(b) 16.2	(c) 13.9	(e) -10.0	(f) 75.9	(g) 3.8	(h) 1.9	(i) -12.8	(j) 77.2	(k) -15.0	(l) 78.3	(m) -17.2	(n) 79.3	(o) -20.0	(p) 80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	14.1	20.9	33.7	49.6	62.2	70.5	75.1	71.8	62.1	47.9	30.3	18.0		
	DBStd	22.75	7.69	8.93	9.75	9.27	7.67	6.11	4.78	5.06	6.75	8.41	9.88	7.87		
	HDD50	4290	1112	813	512	118	6	0	0	0	5	139	593	992		
	HDD65	7574	1577	1233	970	467	145	19	1	6	129	530	1040	1457		
	CDD50	3008	0	0	7	106	383	614	777	675	369	75	2	0		
	CDD65	820	0	0	0	5	57	184	313	217	43	1	0	0		
	CDH74	7737	0	0	2	95	742	1748	2811	1838	476	25	0	0		
	CDH80	2666	0	0	0	20	232	634	1085	593	100	2	0	0		
	WSAvg	5.3	5.0	5.4	6.6	7.2	6.8	5.1	4.4	4.1	4.5	5.0	5.0	5.1		
	PrecAvg	14.9	0.0	0.1	0.3	0.7	1.5	2.8	4.7	2.4	1.3	0.8	0.3	0.0		
(15) Precipitation	PrecMax	20.6	0.3	0.4	1.0	2.2	3.5	5.4	10.8	5.7	3.5	2.6	1.7	0.2		
	PrecMin	9.4	0.0	0.0	0.0	0.0	0.1	0.5	0.8	0.5	0.2	0.0	0.0	0.0		
	PrecStd	3.1	0.1	0.1	0.3	0.6	0.9	1.4	2.1	1.2	0.8	0.6	0.4	0.1		
	PrecStd	3.1	0.1	0.1	0.3	0.6	0.9	1.4	2.1	1.2	0.8	0.6	0.4	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	51.1	68.3	82.8	91.4	94.6	96.1	92.5	87.0	77.3	61.9	44.6		
		MCWB	30.4	36.5	46.2	54.4	60.3	63.2	70.7	69.9	64.1	55.9	44.1	34.1		
	2%	DB	34.9	45.8	61.2	76.0	85.9	90.4	92.2	88.7	82.8	71.6	55.3	38.5		
		MCWB	26.3	33.3	42.6	51.9	58.3	64.1	69.8	68.8	61.2	52.4	41.6	29.3		
	5%	DB	31.2	41.4	55.5	71.5	81.8	86.8	89.0	86.1	79.5	67.1	50.3	34.6		
		MCWB	23.7	30.8	39.4	49.7	56.2	62.7	69.2	67.6	59.6	50.3	38.9	26.6		
	10%	DB	27.7	36.9	50.5	67.0	77.7	83.4	86.1	83.3	76.1	62.6	45.7	31.1		
		MCWB	21.1	28.1	36.8	47.6	54.8	62.2	68.4	66.6	58.4	48.1	35.7	24.1		
	0.4%	WB	30.6	38.0	47.1	57.8	63.5	69.8	75.8	75.2	67.9	58.4	45.9	34.8		
		MCDB	39.8	48.5	65.3	75.5	82.2	83.3	88.8	87.3	81.8	71.7	56.3	45.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.5	33.9	43.5	54.0	61.0	68.0	73.8	72.8	64.8	55.0	42.2	29.5		
		MCDB	34.6	45.0	59.6	71.5	79.4	80.9	85.7	83.2	76.3	67.5	54.0	38.0		
	5%	WB	23.8	31.0	40.2	51.2	58.9	66.6	72.4	71.0	62.6	52.3	39.2	26.8		
		MCDB	31.0	41.3	54.2	68.0	75.4	78.8	83.2	80.7	74.9	64.2	49.7	34.4		
	10%	WB	21.2	28.1	37.1	48.7	56.9	65.2	70.9	69.4	60.5	49.6	35.9	24.2		
		MCDB	27.5	36.9	50.2	65.5	73.6	77.2	81.0	78.4	72.1	60.7	45.3	30.9		
	5% DB	MDBR	20.7	22.0	21.9	22.6	22.4	19.8	18.3	19.3	22.5	21.9	20.0	19.6		
		MCDBR	24.6	27.5	29.1	29.2	28.7	25.5	22.9	23.3	27.5	27.8	25.7	23.3		
		MCWBR	18.2	18.5	16.2	13.4	10.7	8.2	7.2	8.2	10.2	13.3	15.5	16.5		
		MCWBR	24.5	26.6	27.6	25.5	23.3	19.7	18.6	18.7	21.8	24.1	24.7	23.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	18.3	18.2	15.6	13.0	10.1	7.4	7.4	7.9	9.6	12.8	15.4	16.6		
		MCWBR	18.3	18.2	15.6	13.0	10.1	7.4	7.4	7.9	9.6	12.8	15.4	16.6		
	Clear-Sky Solar Irradiance	taub	0.291	0.344	0.392	0.465	0.527	0.584	0.546	0.501	0.451	0.404	0.329	0.298		
		taud	2.298	2.138	2.035	1.895	1.798	1.751	1.904	1.996	2.040	2.072	2.224	2.266		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	268	266	266	255	242	228	236	242	245	240	248	252		
		Edh at Noon	27	38	48	59	67	71	60	53	47	40	29	26		
(44) All-Sky Solar Radiation	RadStd	RadAvg	802	1112	1489	1772	1960	1817	1770	1712	1483	1136	810	686		
		RadStd	46	68	93	109	127	142	105	98	79	69	58	37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.25	N/A	-0.65	-0.90	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	-2.36	N/A	N/A	N/A	N/A	N/A	+103	N/A

Nomenclature: See separate page